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Author(s): Marjorie Lee Browne

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A NOTE ON THE CLASSICAL GROUPS

MARJORIE LEE BROWNE, North Carolina College at Durham

The purpose of this paper is to set forth some topological properties of and relations between certain classical groups. While much of the material included here may be known to a few, the main interest of this paper lies in the simplicity of the proofs of some important, though obscured, results. The groups considered are the full linear group of $n \times n$ matrices over the complex numbers, denoted by $GL(n, c)$, and some of its subgroups: $GL(n, r)$, the subset having real coefficients; $SL(n, c)$, the subset of $GL(n, r)$ with determinant unity; $U(n)$, the set of unitary matrices; $SU(n)$, the unimodular group; $O(n)$, the real orthogonal group; $SO(n)$, the rotation group; and $GL(n, r)^+$, the elements of $GL(n, r)$ having positive determinants. The symbolic notions and definitions of the groups mentioned above are the same as used in [1].

1. Decompositions as topological spaces. It has been shown in [1] that the underlying space of $GL(n, c)$ is the topological product of the space of the n -dimensional unitary group and that of the positive definite hermitian matrices. A second decomposition of this space gives $GL(n, c)$ as the topological product of the non-zero complex numbers and the space of $SL(n, c)$, the commutator subgroup of the algebraic group $GL(n, c)$ and a subgroup of the topological group $GL(n, c)$. Consider the matrix $A \in GL(n, c)$ and define the matrix $P = (p_{ij})$ such that $p_{11} = |A|$, $p_{ii} = 1$ for $i \neq 1$, and $p_{ij} = 0$, $i \neq j$. Also define $B = (b_{ij})$ by letting $b_{1j} = a_{1j}/|A|$ and $b_{ij} = a_{ij}$ for $i \neq 1$. Clearly $PB = A$ and $|B| = 1$. Since P and B are continuous functions of A , the decomposition is continuous. The intersection of all matrices of type P and all matrices of type B contains only the identity, and the decomposition is therefore unique. Since multiplication of matrices is continuous, the identification is a homeomorphism. The set of all matrices of type P is homeomorphic to the non-zero complex numbers.

Simple considerations of the above decompositions of $GL(n, c)$ will yield two decompositions of the underlying topological space of $GL(n, r)$. First, if $\sigma \in GL(n, r)$, then $\sigma = \tau\alpha$, where τ is unitary, α a positive definite hermitian matrix, and $\sigma = \bar{\sigma}$. Since the decomposition of σ , as an element of $GL(n, c)$, is unique; $\tau\alpha = \bar{\tau}\bar{\alpha}$ implies $\tau = \bar{\tau}$ and $\alpha = \bar{\alpha}$; therefore τ is orthogonal and α is positive definite symmetric. Thus, it is proved that the underlying topological space of $GL(n, r)$ is the topological product of that of $O(n)$ and the space of positive definite symmetric matrices. Secondly, let $A \in GL(n, r)$, then $A = PB$ where P and B are defined as above, and $B \in SL(n, c)$. Similarly, $A = \bar{A}$ and the uniqueness of the second decomposition in the preceding paragraph implies that P is real, $B \in SL(n, r)$, and that the underlying space of $GL(n, r)$ is the topological product of the non-zero reals and the underlying space of $SL(n, r)$. A third decomposition, due to the Gram-Schmidt orthogonalization process, gives $GL(n, r)$ as the product of the space of $O(n)$ and that of the triangular matrices with $a_{ii} > 0$.

It follows easily that the space of $GL(n, r)^+$, the component of the identity in $GL(n, r)$, can be decomposed into (1) the product of $SO(n)$, the n -dimensional rotation group, and the set of positive definite symmetric matrices (2) the product of the positive reals and $SL(n, r)$, and (3) the space of $SO(n)$ and the triangular matrices with $a_{ii} > 0$.

Let τ be an element of $SL(n, r)$, then $\tau = \sigma\alpha$; σ is orthogonal and α is positive definite symmetric, since τ is also an element of $GL(n, r)$. Since the determinant of τ equals one, σ and α are of determinant one and the uniqueness of the decomposition in $GL(n, r)$ proves that $SL(n, r)$ is the topological product of the underlying space of $SO(n)$ and that of the positive definite symmetric matrices of determinant one. A similar argument shows that this space, $SL(n, r)$, is also the topological product of $SO(n)$ and the space of all triangular matrices having determinant one and main diagonal elements positive. That $SL(n, c)$ can be decomposed into the product of $SU(n)$ and the space of positive definite hermitian matrices of determinant one can be proved easily using the first decomposition of $GL(n, c)$. Moreover, $U(n)$ is shown to be the product of $SU(n)$ and the complex numbers of absolute value one in [1].

2. Connectivity. It is well known that many of the underlying spaces are connected [1]. But by proving $SU(n)$ is connected by first principles, the connectivity of the others comes easily through a consideration of their decompositions and an elementary property of projections. Any $S \in SU(n)$ can be written as UBU^{-1} with U unitary and B a diagonal matrix having $b_{ii} = e^{i\theta_i}$. Define B_t to be the diagonal matrix formed from B by letting $b_{ii} = e^{it\theta_i}$ for $0 \leq t \leq 1$. It is clear that $UB_tU^{-1} = f(t)$ is a curve from B to 1. And $SU(n)$ is therefore connected. Since $U(n) = T^1 \times SU(n)$, and T^1 is homeomorphic to the unit circle, $U(n)$ is connected. The set of positive definite hermitian matrices is homeomorphic to a Euclidean space [1]. This fact together with the connectivity of $U(n)$ and the first decomposition of $GL(n, c)$ proves that $GL(n, c)$ is connected. Since the projection function of a product onto one of its factors is continuous, $SL(n, c)$ is a connected set (consider the second decomposition of $GL(n, c)$). $GL(n, r)^+$ is a connected set, because it is the component of 1 in $GL(n, r)$. $SO(n)$ is the image of this connected set under the projection function on the first decomposition of $GL(n, r)^+$, and $SO(n)$ is therefore connected. Similarly $SL(n, r)$, being the image of this function on the second decomposition of $GL(n, r)^+$, is a connected set.

3. Deformation retracts. A space X is a deformation retract of Y relative to X if there exists a homotopy f of Y into X such that (1) $f(0, y) = y$, (2) $f(1, y) \in X$ for all $y \in Y$, and (3) $f(t, x) = x$ for all $x \in X$.

The following considerations will show that $SO(n)$, $O(n)$ and $U(n)$ are deformation retracts of $GL(n, r)^+$, $GL(n, r)$, and $GL(n, c)$ respectively. The mapping

$$f(t, \tau) = \sigma(t1 + (1 - t)\alpha) \quad 0 \leq t \leq 1$$

is clearly a homotopy of $GL(n, r)^+$, $GL(n, r)$ and $GL(n, c)$, where σ is an element of $SO(n)$, $O(n)$, and $U(n)$ respectively and α is a positive definite symmetric matrix if $\tau \in GL(n, r)^+$ and if $\tau \in GL(n, r)$, and a positive definite hermitian matrix if $\tau \in GL(n, c)$ (see first decompositions of these groups in section 1). Clearly $t1 + (1-t)\alpha$ is symmetric (hermitian) according as α is symmetric (hermitian), since each set of matrices is closed with respect to scalar multiplication and addition. Moreover, if $\{\lambda_{ii}\}$ are eigenvalues of α , then $\{t + (1-t)\lambda_{ii}\}$ are eigenvalues of $t1 + (1-t)\alpha$. Moreover for any t these eigenvalues are positive, if the λ_{ii} are positive. Finally if $\tau \in GL(n, r)^+$, $GL(n, r)$ and $GL(n, c)$, respectively, then $f(0, \tau) = \tau$, $f(1, \tau) = \sigma$, and $f(t, \sigma) = \sigma$ for all t .

Secondly, consider the mapping

$$f(t, A) = \frac{1}{(\sqrt[n]{|A|})^t} A, \quad 0 \leq t \leq 1,$$

on each of the following sets: $GL(n, r)^+$, the set of positive definite symmetric matrices, the set of positive definite hermitian matrices, and the set of real triangular matrices $A = (a_{ij})$ with $a_{ii} > 0$. It is clearly a homotopy of these sets. Moreover, using this homotopy one proves easily that $SL(n, r)$, the set of positive definite symmetric matrices of determinant one, the set of positive definite hermitian matrices of determinant one and the set of real triangular matrices $A = (a_{ij})$, $a_{ii} > 0$ and $|A| = 1$ are respectively deformation retracts of the above named sets.

Finally, let τ be an element of $SL(n, r)$, then by the second decomposition of this space, $\tau = \sigma\alpha$, where σ is an element of $SO(n)$ and $\alpha = (a_{ij})$, a triangular matrix having $a_{ii} > 0$ and $|\alpha| = 1$. Define $\alpha_t = (b_{ij})$, such that $b_{ii} = a_{ii}^t$, and $b_{ij} = ta_{ij}$ for $i \neq j$ and $0 \leq t \leq 1$. The mapping

$$f(t, \tau) = \sigma\alpha_t$$

is clearly a homotopy of $SL(n, r)$. Moreover, it is easily seen that $f(0, \tau) = \sigma$, $f(1, \tau) = \tau$ and $f(t, \sigma) = \sigma$ for each t . Therefore $SO(n)$ is a deformation retract of $SL(n, r)$.

4. Homotopy types. It is well known that if Y_0 is a deformation retract of Y relative to Y_0 , then Y_0 and Y are of the same homotopy type [3]. From the preceding considerations it follows that the following sets of spaces are of the same homotopy type:

- (1) $U(n)$ and $GL(n, c)$
- (2) $O(n)$ and $GL(n, r)$
- (3) $SO(n)$, $GL(n, r)^+$ and $SL(n, r)$
- (4) The set of positive definite symmetric (hermitian) matrices and the set of positive definite symmetric (hermitian) matrices of determinant one.
- (5) The set of real triangular matrices with positive diagonal terms and its subset of matrices with determinant one.

It might be observed at this point that the set of positive definite symmetric matrices and the set of real triangular matrices with positive diagonal terms are homeomorphic, and hence are of the same homotopy type. For each of these sets is homeomorphic to $R^{n(n+1)/2}$. One notes that each is homeomorphic to $R^{n(n-1)/2} \times R^n(+)$, where $R^n(+)$ is that octant of R^n with all coordinates positive; this set is clearly homeomorphic to R^n .

Finally, since homeomorphic sets are of the same homotopy type and since this last relation is one of equivalence, the sets of symmetric matrices in (4) and the spaces in (5) are of the same homotopy type.

5. Further considerations of $SL(n, c)$, $U(n)$ and $SU(n)$. This paper does not establish $SU(n)$ as a deformation retract of $SL(n, c)$. But one observes that their n -th homotopy groups are isomorphic. The set of positive definite hermitian matrices of determinant one is homeomorphic to R^{n^2-1} . Therefore $SL(n, c) = R^{n^2-1} \times SU(n)$. It has been shown elsewhere [2], that if X_1, X_2 , and X_3 are topological spaces and if $X_1 = X_2 \times X_3$, then $\pi_n(X_1) = \pi_n(X_2) \times \pi_n(X_3)$, where $\pi_n(X_i)$ are n -th homotopy groups of X_i . Also since any map into R^k is homotopic to the identity, $\pi_n(R^k) = I$ for all n and any k , and $\pi_m[SL(n, c)] = \pi_m[SU(n)]$ for all m . Since the m -th homotopy groups of the non-zero complex numbers and of the complex numbers of absolute value one contain only the identity for $m > 1$, it follows that $\pi_m[GL(n, c)] = \pi_m[SL(n, c)] = \pi_m[SU(n)] = \pi_m[U(n)]$, for all $m > 1$. Moreover, $\pi_m[GL(n, c)] = \pi_m[U(n)]$ for all m .

6. Further consideration of $GL(n, r)$, $O(n)$, $GL(n, r)^+$, $SL(n, r)$, $SO(n)$. The first two of these groups are not connected sets [1], but they are homogeneous spaces with $GL(n, r)^+$ and $SO(n)$ as components of their identity elements respectively. In [2, 114], one observes that $\pi_m[O(n)] = \pi_m(SO(n))$. Therefore these five groups have isomorphic m -th homotopy groups for $m \geq 1$, since the first two groups are of the same homotopy type and the last three are of the same homotopy type. It is known that m -th homotopy groups are invariants of the same homotopy type [2].

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